

Trace

$$\text{Tr}(A) = a_{11} + a_{22} + \dots$$

$\downarrow$
 a square matrix + a nxn matrix

Linear transformation

1. $T(\vec{u} + \vec{v}) = T(\vec{u}) + T(\vec{v})$
T preserves vector addition
2. $T(\alpha \vec{u}) = \alpha T(\vec{u})$
T preserves scalar multiplication

Matrix Inverse

$$A \cdot A^{-1} = I_n$$

$$A^{-1} \cdot A = I_n$$

$$A \vec{e}_j = \vec{e}_j$$

$\downarrow$
 $A^{-1}(\vec{e}_j)$

If A is invertible

$$A\vec{x} = \vec{b} \text{ has unique solution}$$

$$\vec{x} = A^{-1}\vec{b}$$

The matrix of T with respect to the ordered bases B and C

$$[T]_{C,B} = \begin{bmatrix} [T(b_1)]_C \\ [T(b_2)]_C \\ \vdots \end{bmatrix}$$

$$[T]_B = P_{B,C} [T]_C P_{C,B}$$

Rank and nullity theorem

$$\text{Number of columns} = \text{rank}(A) + \text{nullity}(A)$$

Rows of B form basis for row space of A
columns of A that have pivots in B form a basis for column space of A.

Transformation

Rotation by θ

$$\begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}$$

Expansion along x-axis by a factor C

$$\begin{bmatrix} C & 0 \\ 0 & 1 \end{bmatrix}$$

shear in x-direction with factor C

$$\begin{bmatrix} 1 & C \\ 0 & 1 \end{bmatrix}$$

To be a basis
span V $\phi = k_1 \{e_1\} \cup k_2 \{e_2\}$
 $= k_1 \{e_1\} \cup k_2 \{e_2\}$

① the elements in the set must be linearly independent

② number of elements in the set equal to $\dim(V)$
 $\therefore$ Able to span V

Least square

Best approximation to $\vec{y}$ inside the image of A
Sample is the vector

$$\begin{aligned} \begin{bmatrix} a \\ b \end{bmatrix} \vec{u} &= \text{Proj}_{\text{col}(A)}(\vec{y}) \\ &= A(A^T A)^{-1} A^T \vec{y} \\ &= (A^T A)^{-1} A^T \vec{y} \end{aligned}$$

$$\vec{y} = y \vec{u} \quad A = \begin{bmatrix} x_1 & x_2 \\ x_1 & x_2 \\ x_1 & x_2 \end{bmatrix}$$

$y = a + bx$

Find determinant use row operation

row reduce to pivot
以下全是零

$\det(A)$ = 对角线上的数相乘
两行 factor k
if times any row by k
乘 -1

* row 1 $\rightarrow$ row 2 + 2 row 1
if swapped two rows
不 change determinant

Subspace

0. W contains the zero vector $\vec{0}$ from V

1. W is closed under vector addition

$$\vec{u}, \vec{v} \in W \Rightarrow \vec{u} + \vec{v} \in W$$

2. W is closed under scalar multiplication

$$\vec{u} \in W, \alpha \in \mathbb{R} \Rightarrow \alpha \vec{u} \in W$$

With respect to a basis

$$[T(\vec{v})]_C = [T]_{C,B} [\vec{v}]_B$$

$$[x]_B = P_{B,S} [x]_S$$

$$\begin{array}{ccc} \vec{x} & \xrightarrow{\text{乘 } A} & T(\vec{x}) \\ \downarrow \text{乘 } C^{-1} & & \downarrow \text{乘 } C^{-1} \\ [\vec{x}]_B & \xrightarrow{\text{乘 } D} & [T(\vec{x})]_B \end{array}$$

Dot product

complex dot product

$$\begin{aligned} \langle \vec{u}, \vec{v} \rangle &= \vec{u} \cdot \vec{v} \\ &= u_1 \bar{v}_1 + u_2 \bar{v}_2 \dots \end{aligned}$$

length of a complex vector

$$\begin{aligned} \|\vec{v}\| &= \sqrt{\vec{v} \cdot \vec{v}} \\ &= \sqrt{v_1 \bar{v}_1 + v_2 \bar{v}_2 \dots} \end{aligned}$$

determinant property

$$\det(KA) = K^n \det(A)$$

$\downarrow$
 number of rows

$$\det(AB) = \det(A) \det(B)$$

$$\det(A^T) = \det(A)$$

$$\det(A^{-1}) = \frac{1}{\det(A)}$$

Surjective: at least one solution

$$n \geq m = \text{rank}(A)$$

$\downarrow$
 列数

Injective: at most one solution

$$m \geq n = \text{rank}(A)$$

$$\dim(\text{null}(A)) = 0$$

Bijjective: $m = n = \text{rank}(A)$

$$\det(A) \neq 0$$

* if A does not have full rank, A is not injective/surjective
 $\text{rank}(A) < m$

$$D = C^{-1}AC$$

o D is the transformation matrix for T with respect to the basis B

o C is the change of basis for B. $C = \begin{bmatrix} b_1 & b_2 & \dots \end{bmatrix}$

o A is the T matrix with respect to the standard basis

Algebraic multiplicity

$(\lambda - \lambda_0)^n$ in the characteristic polynomial

Geometric multiplicity

number of linearly independent eigenvectors of λ

Diagonalisable if A.m = G.m for every λ eigenvalue

Orthogonal projection of $\vec{x}$ onto a subspace V

$$\text{Proj}_V \vec{x} = A(A^T A)^{-1} A^T \vec{x}$$

If $A = \{\vec{a}_1, \dots, \vec{a}_k\}$ is orthonormal set for V

则

$$\begin{aligned} \text{Proj}_V \vec{x} &= A A^T \vec{x} \\ &= \langle \vec{x}, \vec{a}_1 \rangle \vec{a}_1 + \dots + \langle \vec{x}, \vec{a}_k \rangle \vec{a}_k \end{aligned}$$

The Gram-Schmidt Algorithm

Given a linearly independent set $\{\vec{v}_1, \dots, \vec{v}_k\}$.

Find orthonormal basis $\{\vec{u}_1, \dots, \vec{u}_k\}$

Shortest distance

from $\vec{x}$ to W

subspace span by orthogonal matrix

$$\|\vec{x} - \text{Proj}_W \vec{x}\|$$

$$\text{Proj}_L(\vec{x}) = \frac{\vec{v} \cdot \vec{x}}{\vec{v} \cdot \vec{v}} \vec{v}$$

$L = \text{span}(\vec{v})$
orthogonal projection of a vector $\vec{x}$ onto L
* $\|\vec{v}\| = \sqrt{\vec{v} \cdot \vec{v}}$

Orthogonal matrix

$$Q^{-1} = Q^T$$

$$\det(Q) = 1 \text{ or } -1$$

Orthogonal set

$$\{\vec{v}_1, \dots, \vec{v}_k\}$$

$$\langle \vec{v}_i, \vec{v}_j \rangle = 0 \quad i \neq j$$

Orthonormal set

$$\textcircled{1} \langle \vec{v}_i, \vec{v}_j \rangle = 0, i \neq j$$

$$\textcircled{2} \|\vec{v}_i\| = 1 \text{ 或者 } \langle \vec{v}_i, \vec{v}_i \rangle = 1$$

Find area of triangle

with three point

$$\vec{u} = OA - OB$$

$$\vec{v} = OC - OB$$

$$\text{Area} = \frac{1}{2} \|\vec{u} \times \vec{v}\|$$

Cross product

$$\vec{u} \times \vec{v} = \begin{vmatrix} i & j & k \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix}$$

$\|\vec{u} \times \vec{v}\|$ = Area of the parallelogram

* $\vec{u} \times \vec{v}$ gives a vector perpendicular to both $\vec{u}$ and $\vec{v}$

Inner Product

2. check axiom

Axiom

1. Symmetry

$$\langle \vec{u}, \vec{v} \rangle = \langle \vec{v}, \vec{u} \rangle$$

2. Linearity of scalar multiplication

$$\langle \alpha \vec{u}, \vec{v} \rangle = \alpha \langle \vec{u}, \vec{v} \rangle$$

3. Linearity of vector addition

$$\langle \vec{u} + \vec{v}, \vec{w} \rangle = \langle \vec{u}, \vec{w} \rangle + \langle \vec{v}, \vec{w} \rangle$$

4. positivity

$$a. \langle \vec{u}, \vec{u} \rangle \geq 0$$

$$b. \langle \vec{u}, \vec{u} \rangle = 0 \Rightarrow \vec{u} = \vec{0}$$

Inner over $\mathbb{C}$

Axiom

1. Hermitian symmetry

$$\langle \vec{u}, \vec{v} \rangle = \overline{\langle \vec{v}, \vec{u} \rangle}$$

2. Linearity of scalar multiplication

$$\langle \alpha \vec{u}, \vec{v} \rangle = \alpha \langle \vec{u}, \vec{v} \rangle$$

3. Linearity of vector addition

$$\langle \vec{u} + \vec{v}, \vec{w} \rangle = \langle \vec{u}, \vec{w} \rangle + \langle \vec{v}, \vec{w} \rangle$$

4. positivity same as left.

Length of $\vec{u}$

$$\sqrt{\langle \vec{u}, \vec{u} \rangle}$$

Distance between $\vec{u}$ and $\vec{v}$

$$\|\vec{u} - \vec{v}\| = \sqrt{\langle \vec{u} - \vec{v}, \vec{u} - \vec{v} \rangle}$$

angle between $\vec{u}$ and $\vec{v}$

$$\cos(\theta) = \frac{\langle \vec{u}, \vec{v} \rangle}{\|\vec{u}\| \|\vec{v}\|}$$

Cartesian equa: $ax+by+cz=d$

Parametric equa: $x=4+2s-3t$
 $y=s$
 $z=t$

Vector equa: $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} a \\ b \\ c \end{bmatrix} + s \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} + t \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$

Other Cartesian:

$$\frac{x-4}{1} = \frac{y-2}{-3} = \frac{z}{5}$$

$$t = \frac{x-4}{1} \quad x = t+4$$

$$t = \frac{y-2}{-3} \quad y = -3t+2$$

→ Cartesian equa for 3 points plane

① Find $\vec{AB}$ and $\vec{AC}$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = s \vec{AB} + t \vec{AC}$$

② make s, t the subject

$$\text{sub A into } ax+by+cz=d$$

make s, t the subject, then

make them into one equa

$$ax+by+cz=d$$

sub 3th A into d

→ angle between two cartesian equa

$$\vec{u} = x-y+z = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

$$\vec{v} = -x+4y-z = 3 \quad \vec{v} = \begin{bmatrix} -1 \\ 4 \\ -1 \end{bmatrix}$$

$$\text{angle } \cos(\theta) = \frac{\langle \vec{u}, \vec{v} \rangle}{\|\vec{u}\| \|\vec{v}\|}$$

→ Volume for OA, OB, OC

$$\text{Volume} = a \cdot (\vec{b} \times \vec{c})$$

$$= \det \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix}$$

→ Cartesian equa for plane pass point and a plane

sub 两个 t 值进 plane 的 equa 得出 2 个 point 再用 3 个 points 求 equa.

Diagonalisable { 3 distinct eigenvalue
columns are $\vec{v}_1, \dots, \vec{v}_3$
orthonormal eigenvectors
Unitary matrix: $U^{-1} = U^T$
Hermitian matrix
 $A = UDU^{-1} = UDU^T$

Hermitian matrix

$$A = UDU^{-1} = UDU^T$$